

ANTHONY BWEMBYA

Experimental Physicist | Detector Systems | Signal Processing | Scientific Computing

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PROFILE

Experimental physicist with a PhD in Experimental Physics from Radboud University with more than five years of experience in detector systems, sensor calibration, scientific instrumentation, signal processing, algorithm development, and experimental data analysis. My doctoral work within the Pierre Auger Observatory centres on the AugerPrime Radio Detector and the development of radio interferometric reconstruction methods for a large distributed sensor array. I combine hands on detector work with Python and C++ software, simulation, numerical optimisation, uncertainty analysis, and system validation. Additional experience includes radiation detection, LiDAR and fluorescence camera calibration, PMTs, CVD diamond sensors, VNA and DAC/AWG instrumentation, and field operation of complex experimental systems.

CORE EXPERTISE

- Experimental physics and complex sensor systems; detector characterisation; calibration; instrumentation; system validation
- RF sensing and signal processing; radio interferometry; antenna response; waveform analysis; timing synchronisation
- Python and C++; Linux; scientific software; automated measurement workflows; data quality diagnostics
- Simulation and modelling; CORSIKA and CoREAS; numerical optimisation; uncertainty propagation; model to data comparison
- Radiation detection and nuclear instrumentation; spectroscopy; X ray fluorescence; detector calibration and measurement traceability
- Optical and imaging instrumentation exposure including LiDAR, fluorescence cameras and PMTs; laboratory and field measurements
- Cross functional work with physicists, engineers, software developers and field technicians in international collaborations

PROFESSIONAL AND RESEARCH EXPERIENCE

PhD Researcher, Experimental Detector Physics and Radio Interferometry

Radboud University / Pierre Auger Observatory, Netherlands and Argentina | 2021 to 2026

- Developed precision timing and calibration methods for the AugerPrime Radio Detector, a geographically distributed antenna array operating in demanding environmental conditions.
- Developed and validated an automated timing calibration workflow using stable radio transmitters and independent validation signals. In a representative event, reduced failing triangle closure tests from 334 to zero and improved the average closure residual from approximately 25.4 ns to 0.18 ns.
- Developed and evaluated radio interferometric reconstruction algorithms for highly inclined cosmic ray air showers, combining signals from spatially separated sensors under timing, geometry, noise and atmospheric constraints.
- Built reproducible Python and C++ pipelines for signal extraction, phase analysis, detector quality diagnostics, noise characterisation, simulation and reconstruction.
- Used numerical optimisation methods including L BFGS B, Powell and differential evolution to improve reconstruction and resolve calibration ambiguities.
- Investigated antenna response, detector geometry, atmospheric refractivity, timing uncertainty and amplitude systematics through controlled simulation and sensitivity studies.
- Supported LiDAR and fluorescence camera calibration and operations for atmospheric monitoring and detector performance assessment.
- Worked with scientists, engineers, software developers and field technicians across an international experimental collaboration; documented algorithms and validation procedures and communicated results at collaboration meetings and workshops.

Radiation Protection Officer

Radiation Protection Authority of Zambia | 2021 to 2022

- Performed radiation protection assessments and technical investigations using radiation measurement data, calibration records and uncertainty aware interpretation.
- Reviewed detector calibration and measurement traceability for laboratories and regulated organisations and prepared technical reports and recommendations.

Research and Laboratory Intern

National Institute for Scientific and Industrial Research, Zambia

- Worked with alpha and gamma spectroscopy, X ray fluorescence, detector based measurements, calibration, sample preparation and non destructive radiation analysis.

SELECTED INSTRUMENTATION AND DETECTOR EXPERIENCE

- AugerPrime Radio Detector: distributed radio sensor array, timing calibration, antenna response, signal processing and interferometric reconstruction
- Belle II / JENNIFER2: CVD diamond sensor work connected with LUMIBELLE2 detector instrumentation
- Optical and atmospheric instrumentation: LiDAR operation, fluorescence camera calibration and PMT calibration
- Laboratory tools: VNA, DAC/AWG, radiation detectors, spectroscopy systems and IV/CV related measurement familiarity
- Field systems: detector commissioning, remote monitoring, sensor diagnostics and measurements in international experimental environments

EDUCATION

PhD in Experimental Physics Radboud University, Netherlands | 2021 to 2026

Thesis: In Phase With the Cosmos: Mass Composition of Cosmic Rays via Radio Interferometry. Defended 6 October 2026.

MSc in Nuclear Physics and Technology National Research Nuclear University MEPhI, Russia | 2018 to 2021

Research in the Department of Experimental Nuclear Physics and Cosmophysics; GALPROP modelling and PAMELA / AMS 02 data analysis.

BSc in Physics University of Zambia | 2017

TECHNICAL TOOLKIT

Programming: Python, C++, Bash, Linux, Git **Scientific:** NumPy, SciPy, pandas, ROOT, Jupyter, scikit learn, PyTorch

Methods: signal processing, Fourier and spectral analysis, optimisation, simulation, uncertainty propagation, machine learning

LANGUAGES

English: fluent and preferred | Lozi, Bemba, Nyanja: native | Russian: working proficiency | Dutch: developing